

EDUCATION

Doctor of Philosophy in Biology

2026-Present

Saint Louis University, USA

Master of Science in Sustainable Agriculture

2022-2024

GPA: 4.0 *Lincoln University of Missouri, USA*

Thesis title: Integrating Living Perennial Cover Crops into Organic Vegetable Systems: Impacts on Soil Health and Cover Crop Biomass Production

Bachelor of Science (Honors) in Agriculture

2016-2020

GPA: 3.82, Honor Roll Student. *Purbanchal University, Nepal*

Thesis title: Effect of Maleic Hydrazide and Gibberellic Acid on Growth and Yield of African Marigold (*Tagetes erecta* cv. Calcuttia Orange)

PROFESSIONAL AND RESEARCH EXPERIENCE

Graduate Teaching Assistant

August 2026- Present

Department of Biology, Saint Louis University

- Conduct research in soil microbial ecology to determine how agricultural management practices and environmental changes influence soil microbial communities, nitrogen and carbon cycling, greenhouse-gas emissions, and overall soil health.
- Design and carry out field, greenhouse, and laboratory experiments; develop sampling plans; collect and process soil samples; and maintain accurate experimental records and research documentation.
- Measure microbial and biogeochemical processes associated with nutrient cycling, soil organic matter dynamics, carbon sequestration, and greenhouse-gas production.
- Perform molecular laboratory procedures, including soil DNA extraction, quantitative PCR, sample preparation, and amplicon sequencing workflows.
- Apply bioinformatics and computational methods to process, organize, analyze, and interpret microbial community and environmental datasets.
- Prepare scientific manuscripts, research reports, conference abstracts, presentations, figures, and other materials for communicating research outcomes.
- Prepare laboratory materials, equipment, cultures, reagents, and instructional resources for undergraduate biology courses.
- Guide students through microbiology and microbial ecology experiments, explain laboratory methods and scientific concepts, evaluate assignments and laboratory reports, and provide constructive academic feedback.

Soil Health Research Specialist

July 2024- July 2026

Department of Agriculture and Environmental Sciences, Lincoln University of Missouri.

PROJECT 1: Evaluating Climate Resilient Alternative Winter-Feeding Strategy for Improved Livestock Productivity by Grazing in a Solar Corridor Cropping System.

PROJECT 2: Cover Cropping and Crop Rotation Strategies in Organic Vegetable Production Systems to Build Soil Health and Improve Yield and Profitability.

PROJECT 3: Harnessing biochar-biologicals-compost interactions to maximize productivity and sustainability of small- and medium-sized transitioning vegetable production systems.

- Conduct C and N cycling assays, including extracellular enzymes, mineralizable C, POXC, and ACE protein, linking microbial function to nutrient dynamics.

- Apply statistical modelling (R, SAS) to evaluate treatment effects on soil health indicators.
- Quantify aboveground biomass and assess plant tissue nutrient composition (C/N ratio, crude protein and total digestible nutrients) to integrate plant nutrient cycling with soil microbial function.
- Analyze microbial community structure using QIIME2, linking patterns to soil metabolic processes.
- Plan and coordinate field trial operations, including planting, harvesting, intercultural operations, and process streamlining to improve efficiency across diverse field environments.
- Lead the soil sampling effort for field trials, ensuring a systematic and accurate collection of soil samples across diverse locations and conditions.
- Collaborate with researchers, extension specialists, and stakeholders to translate findings into applied recommendations.

Graduate Research Assistant

August 2022- June 2024

Department of Agriculture and Environmental Sciences, Lincoln University of Missouri.

PROJECT 1: Integrating Living Perennial Cover Crops into Organic Vegetable Production System: Impacts on Soil Health and Cover Crop Biomass Production.

PROJECT 2: Soil Health Characteristics Across a Gradient of Organic Land-use Intensity in Mid-Missouri.

- Planned, organized, and assisted in executing multiple field trial activities, including assisting with planting, harvesting, cover crop biomass collection, soil sampling and ensuring high-quality data collection.
- Conducted laboratory procedures on soil health analysis using methods and techniques to analyze soil organic carbon, microbial activity, and nutrient cycling.
- Applied methods of data analysis to interpret results, providing insights into soil health and implications.
- Presented research findings to farmers across the Mid-Missouri area through conferences and extension projects.

Research Technician

June 2021- July 2022

Kisan Care Pvt. Ltd.

- Supported the dissemination of ecological farming techniques such as bio-fertilizers, cover cropping, legume intercropping, and mixed cropping, to commercial vegetable growers.
- Conducted field-level training for over 500 farmers in the Kathmandu valley, equipping them with practical skills in sustainable agriculture.
- Collaborated with senior staff and members of the Kisan Care team to draft various reports, correspondence, and documentation, summarizing periodic findings and the main insights into the project.

Research Intern

August 2020- January 2021

Floriculture Department Center

- Investigated the impact of Maleic Hydrazide and Gibberellic Acid on the growth and productivity of African marigold in Godawari, Nepal.
- Communicated findings and implications for marigold cultivation, offering valuable guidance for farmers interested in enhancing productivity through targeted growth regulator applications.

PUBLICATIONS

Kwarteng, A. O., **Karki, P.**, Hurisso, T. T., Afari, K., & Reinbott, T. M. (2026). Weed Seedbank Response to Cover Crops in Traffic Pathways Between Vegetable Production Beds Under Organic Management in the US Midwest. *Weed Science*, 1–28. doi:[10.1017/wsc.2026.10136](https://doi.org/10.1017/wsc.2026.10136)

Karki P., Hurisso T., Reinbott T. (2026). Soil health responses to annual versus perennial cover crops in alleyways during vegetable-based organic transition (Under review)

Karki P., Hurisso T., Zhang K., Reinbott T. (2026). The Effects of Perennial and Annual Cover Crops on the Structure of Soil Microbial Communities during Transition to Organic Vegetable Production. (Under review)

Karki P., Atreya P.N., Shrestha S. (2021). Effect of Maleic Hydrazide and Gibberellic Acid on growth and yield of African marigold (*Tagetes erecta* L.) cv. Calcuttia Orange. *Fundam Appl Agric* 6(3): 272–278, 2021 doi: [10.5455/faa.103177](https://doi.org/10.5455/faa.103177)

PRESENTATIONS

Hurisso T., Reinbott T., Kremer R., Alsunuse B., Al-Awaal N., **Karki P.**, Afari K. Seasonal Soil Health Dynamics in Solar Corridor Cropping System Under Organic Management. Poster presented at 81st Soil and Water Conservation Society (SWCS) International Annual Conference. 26th July 2026 – 29th July 2026, St. Louis, MO, USA

Swystun E., Francis K., **Karki P.** Deletion of Wsp Gene Global Metabolite Production in *P. aeruginosa* PA 14. Poster presented at 5th International School on Non-target Metabolomics for Natural Products and Microbiome Research. 15th June 2026- 21st June 2026 Denver, CO, USA

Karki P. Soil health comparison of annual versus perennial cover crops in alleyways between vegetable production beds during organic transition. 1890 ARD Research Symposium. 28th March 2026- 1st April 2026, New Orleans, LA, USA

Hurisso T., **Karki P.**, Afari K., Al-Awaal N., Alsunuse B. Do cover crops established in the alleyways impact soil health in adjacent vegetable production beds? Poster presented at 1890 ARD Research Symposium. 28th March 2026- 1st April 2026, New Orleans, LA, USA

Kwarteng A., Hurisso T., **Karki P.**, Al-Awaal A., Afari A. Cover Crops and Weed Seedbank Emergence Dynamics. Poster presented at 1890 ARD Research Symposium. 28th March 2026- 1st April 2026, New Orleans, LA, USA

Karki P., Hurisso T., Kwarteng A., Reinbott T. Perennial versus Annual Cover Crops in Alleyways: Impacts on Soil Health and Weed Suppression in Organic Vegetable Production. Poster presented at 2026 Great Plains Growers Conference. 8th January 2026-10th January 2026, St. Joseph, MO, USA

Karki P., Hurisso T., Reinbott T. Soil health comparison of annual versus perennial cover crops in alleyways between vegetable production beds during organic transition. Poster presented at Soil Health Institute's 10th Anniversary Virtual Meeting. 3rd December 2025- 4th December 2025.

Karki P., Hurisso T., Reinbott T. Assessing the Effect of Living Cover Crops in Alleyways on Soil Health Properties of Adjacent Organic Vegetable Production Rows. Poster presented at 2025 ASA, CSSA, SSSA International Annual Meeting. 9th November 2025- 12th November 2025, Salt Lake City, UT, USA.

Hurisso T., **Karki P.**, Reinbott T. Soil Health Comparison of Living Perennial versus Annual Cover Crops during Transition to Certified Organic Vegetable Production. Poster presented at MarbleSeed Organic Farming Conference, 20th February- 22nd February 2025, La Crosse, WI, USA.

Al-Awwal N., **Karki P.**, Hurisso T. Making Sense of the Soil Health Testing. Minority and Limited Resources Farmers and Producers Conference. 12th September -13th September 2024, Sikeston, MO, USA.

Karki P., Hurisso T., Reinbott T. Integrating Living Perennial Cover Crops into Organic Vegetable Systems: Impact on Soil Health and Cover Crop Biomass Production. Poster presented at 1890 ARD Research Symposium. 7th April – 9th April 2024, Nashville, TN, USA.

Karki P., Hurisso T., Al-Awwal N., Reinbott T., Gerling T. Soil Health and Vegetable Yield Responses to Perennial versus Annual Cover Crops in Traffic Pathways. Poster presented at 2024 Great Plains Growers Conference: 11th January 2024- 13th January 2024, St Joseph, MO, USA.

Karki P., Hurisso T., Al-Awwal N., Reinbott T. Evaluation of Labile C and N Indicators of Soil Health after 2- years of Living Perennial and Annual Cover crops during organic vegetable transition. Poster presented at: 2023 ASA, CSSA, SSSA International Annual Meeting. 29th October 2023- 1st November 2023, St. Louis, MO USA.

Al-Awwal N., Adeyeye A., **Karki P.**, Tesfaye S., Hurisso T. Soil Health Characteristics across a Gradient of Organic Land-Use Intensity in Mid-Missouri. Poster presented at: 2023 ASA, CSSA, SSSA International Annual Meeting. 29th October 2023 - 1st November 2023, St. Louis, MO USA.

Al-Awwal N., **Karki P.**, Hurisso T. Making Sense of the Soil Health Testing. Minority and Limited Resources Farmers and Producers Conference. 28th September 2023 -29th September 2023, Sikeston, MO, USA.

Karki P., Atreya P.N., Shrestha S. Effect of Maleic Hydrazide and Gibberellic Acid on growth, and yield of African marigold (*Tagetes erecta* L.) CV. Calcuttia Orange. Poster presented at: 12th National Horticultural Seminar-2021. 4th March 2021 -5th March 2021, Kirtipur Kathmandu.

CORE COMPETENCIES & SKILLS

- **Soil Health Analysis:** Soil organic carbon, total nitrogen, mineralizable carbon, potentially mineralizable nitrogen, permanganate oxidizable carbon, ACE soil protein, phospholipid fatty acids, pH, electrical conductivity, and nitrate-nitrogen.
- **Soil Enzyme Analysis:** Dehydrogenase, beta-glucosidase, beta-N-acetylglucosaminidase, arylsulfatase, and acid phosphatase.
- **Microbial Ecology and Molecular Methods:** Soil DNA extraction, quantitative PCR, 16S/ITS amplicon sequencing, microbial community profiling, metabolomics, and transcriptomics.
- **Bioinformatics:** QIIME 2, alpha- and beta-diversity analysis, taxonomic classification, relative-abundance analysis, and sequence-data processing.
- **Statistical Analysis and Programming:** R, SAS, Python, GenStat, Minitab, Tableau, mixed-effects models, multivariate analysis, regression, clustering, PERMANOVA, random forest, and structural equation modeling.
- **Experimental Research:** Field, greenhouse, laboratory, and on-farm experimental design; soil sampling; data collection; quality control; and research data management.
- **GIS and Scientific Communication:** ArcGIS, QGIS, proposal writing, manuscripts, technical reports, conference presentations, and preparation of publication-quality figures and tables.

KEY ACCOMPLISHMENTS

- First position in “Evaluation of Labile C and N Indicators of Soil Health after 2-years of Living Perennial and Annual Cover crops during organic vegetable transition” Poster Competition organized by Organic Management Systems Community, ASA-CSSA-SSSA. (\$300) 2023
- Third position in “Integrating Living Perennial Cover Crops into Organic Vegetable Systems: Impact on Soil Health and Cover Crop Biomass Production” Poster Competition organized by 1890 ARD Research Symposium. (\$100) 2024
- Third position in “Soil health comparison of annual versus perennial cover crops in alleyways between vegetable production beds during organic transition” Poster Competition organized by Soil Health Institute. (\$100) 2025
- Third position in 2026 World Technology Partners - Green Sustainability Trends Competition. (\$150) 2026
- Full Research Stipend, Lincoln University of Missouri (\$50,000) 2022-2024
- Highest GPA student in undergraduate degree 2016-2020

CERTIFICATION

- 5th International School on Non-target Metabolomics for Natural Products and Microbiome Research. University of Denver 2026.
- Biology meets Programming: Bioinformatics for Beginners authorized by University of California, San Diego and offered through Coursera 2025.
- Fundamentals of GIS authorized by University of California, Davis and offered through Coursera 2021.
- Sustainable Agricultural Land Management authorized by University of Florida and offered through Coursera 2020.